SOUHAIL MESBAH

Digital Sciences Engineering Student

Contact Info

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✆ 07 46 68 98 92

📍 Toulouse, France

🔗 Git Hudson / [Portfolio Link]

Technical Skills

Python | Ada | MATLAB | Git (Hudson) | CoqIDE

Environments: Windows 11, NVIDIA Hardware, Matrix Computation.

Soft Skills

Complex problem-solving

Mathematical rigor & logic

Adaptability

Languages

Arabic: Native

French: Fluent (Academic)

English: Proficient (Technical)

Japanese: Beginner (Hiragana)

Interests & Hobbies

Table Tennis: Intensive practice (5-10 hrs/week).

Fitness: Regular strength training (2x/week)

> Seeking a 3-month internship in software development, algorithmics, and applied mathematics.

EDUCATION

Engineering Degree in Digital Sciences *2026 – 2029 | INP-ENSEEIHT, Toulouse*

Specialization in software development, algorithmics, and applied mathematics.

Preparatory Classes for Grandes Écoles *2024 – 2026 | Moulay Driss High School, Fès (Morocco)*

MPSI (2024/2025) and MP* (2025/2026) tracks. Intensive preparation in advanced mathematics and engineering sciences.

TECHNICAL PROJECTS

Spatial Detection & Modeling (MATLAB)

- **S** **Situation:** Analysis of noisy spatial data during numerical mathematics practicals.
- **T** **Task:** Numerically estimate the optimal coordinates of a center from a scattered point cloud.
- **A** **Action:** Developed a vectorized maximum likelihood algorithm. Generated uniform random samples to identify the argmax without using time-consuming loops.
- **R** **Result:** Created a highly efficient, fast, and mathematically rigorous script leveraging matrix computation.

Secure Software Development (Ada)

- **S** **Situation:** Creation of an interactive guessing game (`Jeu_Devin`) to learn strict programming principles.
- **T** **Task:** Ensure game reliability and prevent crashes caused by user input errors.
- **A** **Action:** Implemented rigorous input verification to detect if the user entered a string instead of the expected single character.
- **R** **Result:** Completely eliminated runtime bugs linked to erroneous inputs, ensuring a robust codebase.

Logic Verification & Proofs (CoqIDE)

- **S** **Situation:** Introduction to formal methods for software security.
- **T** **Task:** Mathematically prove the validity of algorithms.
- **A** **Action:** Used the Coq proof assistant to write strict